Vineeth Krishna Talasila

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INSPIRE arXiv ORCID

RESEARCH INTERESTS:

Gauge-gravity duality, Quantum Information in Quantum Gravity, Supergravity and String Theory

EDUCATION:

- **Postdoctoral Researcher**, *2024 - present*
  Leinweber Center for Theoretical Physics,
  Department of Physics,
  University of Michigan - Ann Arbor, MI, USA.

- **PhD**,
  Department of Theoretical Physics, *2019 - 2025*
  Tata Institute of Fundamental Research (TIFR), Mumbai, India.
  Thesis Advisor : Dr. Abhijit Gadde
  Thesis Title : "*Multi-partite entanglement in holography*"

- **Integrated BS-MS - Dual degree in Physics and Mathematics**,
  CGPA - 8.83 (on a scale of 10)
  Indian Institute of Science Education and Research (IISER), *2014 - 2019*
  Thiruvananthapuram, Kerala, India.
  Master's Thesis Advisor: Dr. Bindusar Sahoo
  Master's Thesis Title: "*Kaluza-Klein reductions and AdS/RF correspondence*"

RESEARCH PROJECTS:

Leinweber Institute for Theoretical Physics, Michigan *(2024 - Present)*
- Phases of supersymmetric black holes in AdS/CFT
- Gravitational Path Integral
- Super-conformal Indices
- Black Hole decoherence

Tata Institute of Fundamental Research, Mumbai *(2019 - 2024)*
- Supersymmetric Black Holes
- Multipartite entanglement in AdS/CFT (PhD Thesis)
- Non-local defects in Conformal Field Theory (course project)

Indian Institute of Science Education and Research, Thiruvananthapuram *(2018 - 2019)*
- Kaluza-Klein reductions and AdS/RF correspondence (Master's thesis)

LIGO-California Institute of Technology, Pasadena *(May - July 2017)*
- Studying properties of candidate LIGO mirror coatings (Summer Internship)

PUBLICATIONS AND ONGOING PROJECTS:

- V. Krishna, F. Larsen, Allowable complex Black Holes in the Euclidean Gravitational Path Integral, JHEP 06 (2026) 026, [arXiv:2602.05979] [INSPIRE].
- H. Jones, V. Krishna, F. Larsen, Phases of supersymmetric ground states in AdS_4 , JHEP 07 (2026) 174, [arXiv:2510.16274] [INSPIRE].
- S. Choi, D. Jain, S. Kim, V. Krishna, G. Kwon, E. Lee, S. Minwalla, C. Patel, Supersymmetric Grey Galaxies, Dual Dressed Black Holes and the Superconformal Index, SciPost Phys. 19 (2025) 3, 072 [arXiv:2501.17217] [INSPIRE]
- A. Gadde, J. Harper, V. Krishna, Multi-invariants and Bulk Replica Symmetry, JHEP 06 (2025) 116 [arXiv:2411.00935] [INSPIRE]
- S. Choi, D. Jain, S. Kim, V. Krishna, E. Lee, S. Minwalla, C. Patel, Dual Dressed Black Holes as the end point of the Charged Superradiant instability in $\mathcal{N}$=4 Yang Mills, SciPost Phys. 18 (2025) 137 [arXiv:2409.18178] [INSPIRE]
- A. Gadde, S. Jain, V. Krishna, H. Kulkarni and T. Sharma, *Monotonicity conjecture for multi-party entanglement I,* [arXiv:2308.16247] [INSPIRE].
- A. Gadde, V. Krishna, & T. Sharma, Towards a classification of holographic multi-partite entanglement measures. J. High Energ. Phys. 2023, 202 (2023). [arXiv:2304.06082] [INSPIRE].
- A. Gadde, V. Krishna and T. Sharma, *New multipartite entanglement measure and its holographic dual*, Phys. Rev. D 106 (2022) 126001 [arXiv:2206.09723] [INSPIRE].

OTHERS:

- P. Mandal, D. Agarwal, P. Kumar, A. Yelikar, K. Soni and V. Krishna, *Everyday Radio Telescope*, [arXiv:1601.02982].

FELLOWSHIPS and ACHIEVEMENTS:

- Receipient of Kavli-LIGO Caltech Summer Undergrad Research Fellowship ($6600) for an Internship in LIGO laboratory at California Institute of Technology during May - July 2017.
- Receipient of Department of Science and Technology, Government of India INSPIRE Fellowship (Rs 4,00,000 in total) during August 2014 - June 2019.
- Secured All India Rank 6 (number of candidates appeared - 7300 approx.) in Joint Entrance Screening Test (JEST) 2019 for admission to PhD programmes in India.
- Scored All India Rank 19 (number of candidates appeared - 40,000 approx.) in Joint CSIR-UGC National Eligibility Test for admission to PhD programmes in India.

PRESENTATIONS:

TALKS:

- University of Hyderabad, India, Multi-component Black Holes: New Phases of Quantum Gravity, Physics department seminar (November 2026)
- MIT, USA, Multi-component Black Holes: New Phases of Quantum Gravity, Leinweber postdoc meeting (October 2026)
- Virginia Tech, USA, Multipartite Entanglement in Holography, Group seminar (November 2025)
- Kavli-ITS, UCAS, China, New (Non-)Supersymmetric Phases of $\mathcal{N} = 4$ SYM, HET seminar (April 2025)
- IISER, Thiruvananthapuram, India, New measures of multi-partite entanglement and their holographic duals, Department seminar (January 2024)
- IIT Bombay, India, New measures of multi-partite entanglement and their holographic duals, Indian Strings Meeting (December 2023)
- SISSA, Trieste, Italy, Supersymmetric Grey Galaxies and Revolving Black Holes, Department seminar(October 2023)
- University of Milano-Bicocca, Supersymmetric Grey Galaxies and Revolving Black Holes, Department seminar (October 2023)
- ICTP, Trieste, Italy, Supersymmetric Grey Galaxies and Revolving Black Holes, ICTP Workshop on String Theory, Holography and Black Holes - Gong Show talk (October 2023)
- TIFR, Mumbai, Multiboundary wormholes and holographic entanglement, STMP Jouranl Club (June 2022)
- TIFR, Mumbai, New multipartite entanglement measure and it's holographic dual, STMP Journal Club (August 2022)

POSTER:

- (In person) *Quantum Information, Quantum Matter and Quantum Gravity*, YITP, Kyoto, Japan, Supersymmetric Grey Galaxies and Revolving Black Holes (Sep 2023).
- (Online) *It from Qubit,* Perimeter Institute, Towards classification of holographic multi-partite entanglement measures (Aug 2023).
- (In person) *17th Kavli Asian Winter School on Strings, Particles and Cosmology*, Institute for Basic Science, Daejeon, Korea, New multipartite entanglement measure and its holographic dual (Jan 2023).

TEACHING EXPERIENCE:

Teaching Assistant for the following graduate level courses at TIFR, Mumbai:

- Advanced Quantum Mechanics - Course Instructor : Prof. Rishi Sharma (Fall 2020)
- Quantum Field Theory I - Course Instructor : Prof. Suvrat Raju (Spring 2023)
- Quantum Field Theory II - Course Instructor : Dr. Abhijit Gadde (Fall 2023)

TECHNICAL SKILLS:

I have used the following languages/softwares during my research:

Mathematica	Python	C++
MATLAB	SageMath	Inkscape

OUTREACH:

- Organized two workshops on Quantum Black Holes at the Leinweber Institute for Theoretical Physics, University of Michigan.
- Volunteered for an outreach event “Frontiers of Science” at TIFR, Mumbai in 2019 and 2023.
- Volunteered for a Math Circle session 22 conducted by ICTS, Bangalore to help mathematically inclined high school and middle school students in improving their problem solving skills.
- Was a member of Astronomy club of IISER-Thiruvananthapuram from 2014 to 2019 during which we held multiple stargazing sessions.
- Volunteered in Anvesha (Science Fest of IISER, Thirvananthapuram) in 2017.